

Task 1

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python + ~ [ ] [X] ... | [C] X
```

```
PS C:\Users\LENOVO\Desktop\ST2_Week_5_Tutorial_Activities> & C:\Users\LENOVO\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/LENOVO/Desktop/ST2_Week_5_Tutorial_Activities/reursion_mergesort.py  
The sum of the first 5 natural numbers is 15  
The 7th Fibonacci number is 13  
The factorial of 5 is 120  
Insertion Sort took: 0.001713 seconds  
Merge Sort took: 0.001303 seconds  
[2, 6, 9, 5, 3, 4, 5, 6, 6, 7]  
  
[2, 3, 4, 5, 5, 6, 6, 6, 7, 9]  
[1]  
[1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5]  
3240  
PS C:\Users\LENOVO\Desktop\ST2_Week_5_Tutorial_Activities\ST2_Week_5_Tutorial_Activities>
```

```
RecursionError: maximum recursion depth exceeded  
PS C:\Users\LENOVO\Desktop\ST2_Week_5_Tutorial_Activities\ST2_Week_5_Tutorial_Activities> & C:\Users\LENOVO\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/LENOVO/Desktop/ST2_Week_5_Tutorial_Activities/reursion_mergesort.py  
Sum Iterative (n=10): 55  
Fibonacci Iterative (n=10): 55  
Factorial Iterative (n=5): 120  
PS C:\Users\LENOVO\Desktop\ST2_Week_5_Tutorial_Activities\ST2_Week_5_Tutorial_Activities>  
[Previous Line Repeated 991 more times]
```

```
RecursionError: maximum recursion depth exceeded  
PS C:\Users\LENOVO\Desktop\ST2_Week_5_Tutorial_Activities\ST2_Week_5_Tutorial_Activities> & C:\Users\LENOVO\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/LENOVO/Desktop/ST2_Week_5_Tutorial_Activities/reursion_mergesort.py  
  
SUM OF NATURAL NUMBERS  
n=100: Recursive=0.000012s | Iterative=0.000004s  
n=300: Recursive=0.000061s | Iterative=0.000008s  
n=500: Recursive=0.000059s | Iterative=0.000012s  
  
FACTORIAL  
n=50: Recursive=0.000008s | Iterative=0.000006s  
n=200: Recursive=0.000047s | Iterative=0.000013s  
n=500: Recursive=0.000113s | Iterative=0.000052s  
  
FIBONACCI  
n=5: Recursive=0.000004s | Iterative=0.000002s  
n=10: Recursive=0.000010s | Iterative=0.000002s  
n=20: Recursive=0.000013s | Iterative=0.000002s  
PS C:\Users\LENOVO\Desktop\ST2_Week_5_Tutorial_Activities\ST2_Week_5_Tutorial_Activities>
```

Task 2

```
PS C:\Users\LENOVO\Desktop\ST2_Week_5_Tutorial_Activities> & C:\Users\LENOVO\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/LENOVO/Desktop/ST2_Week_5_Tutorial_Activities/reursion_mergesort.py  
The sum of the first 5 natural numbers is 15  
The 7th Fibonacci number is 13  
The factorial of 5 is 120  
Insertion Sort took: 0.001713 seconds  
Merge Sort took: 0.001303 seconds  
[2, 6, 9, 5, 3, 4, 5, 6, 6, 7]  
  
[2, 3, 4, 5, 5, 6, 6, 6, 7, 9]  
[1]  
[1, 1, 1, 1, 1, 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5]  
3240  
PS C:\Users\LENOVO\Desktop\ST2_Week_5_Tutorial_Activities>  
  
PS C:\Users\LENOVO\Desktop\ST2_Week_5_Tutorial_Activities> & C:\Users\LENOVO\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/LENOVO/Desktop/ST2_Week_5_Tutorial_Activities/reursion_mergesort.py  
  
=====
```

```
TESTING DATA SIZE = 100  
=====
```

```
Insertion Sort Time: 0.000126 seconds  
Merge Sort Time:      0.000107 seconds  
  
=====
```

```
TESTING DATA SIZE = 500  
=====
```

```
Insertion Sort Time: 0.004838 seconds  
Merge Sort Time:     0.000718 seconds  
  
=====
```

```
TESTING DATA SIZE = 1000  
=====
```

```
Insertion Sort Time: 0.013606 seconds  
Merge Sort Time:     0.001558 seconds  
  
=====
```

```
TESTING DATA SIZE = 2000  
=====
```

```
Insertion Sort Time: 0.055499 seconds  
Merge Sort Time:     0.002531 seconds  
PS C:\Users\LENOVO\Desktop\ST2_Week_5_Tutorial_Activities>
```

Task 3

```

PS C:\Users\LENOVO\Desktop\ST2_Week_5_Tutorial_Activities\ST2_Week_5_Tutorial_Activities> & C:\Users\LENOVO\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/LENOVO/Desktop/ST2_Week_5_Tutorial_Activities/towerofhanoi.py
Enter number of disks: 4
The moves are:
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Move disk 3 from A to C
Move disk 1 from B to A
Move disk 2 from B to C
Move disk 1 from A to C
Move disk 4 from A to B
Move disk 1 from C to B
Move disk 2 from C to A
Move disk 1 from B to A
Move disk 3 from C to B
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
PS C:\Users\LENOVO\Desktop\ST2_Week_5_Tutorial_Activities\ST2_Week_5_Tutorial_Activities>

```

```

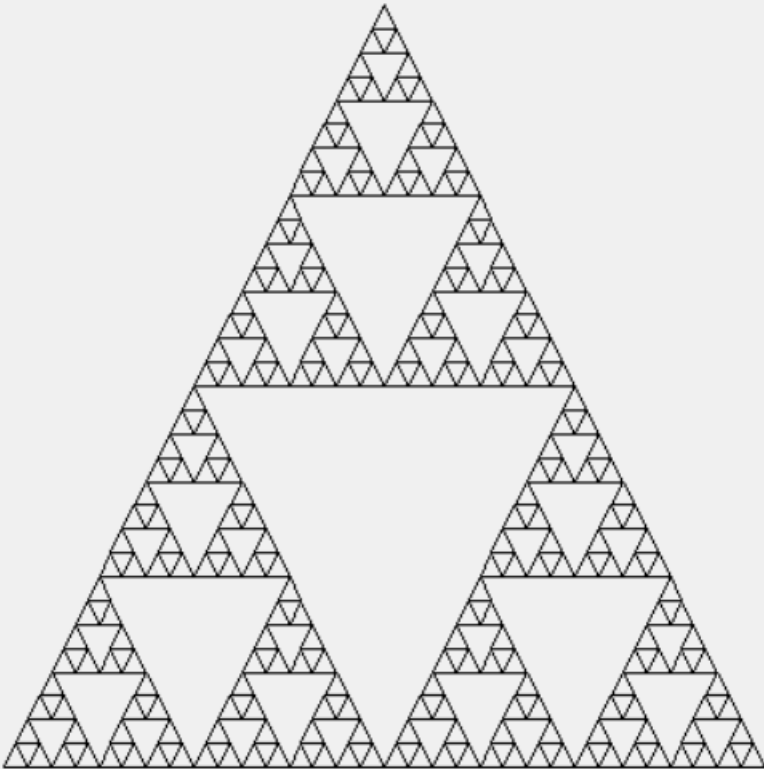
al\Python\pythoncore-3.14-64\python.exe c:/Users/LENOVO/Desktop/ST2_Week_5_Tutorial_Activities/towerofhanoi.py
Enter number of disks: 2
Move disk 1 from A to C
Move disk 2 from A to B
Move disk 1 from C to B
Total number of moves: 3
PS C:\Users\LENOVO\Desktop\ST2_Week_5_Tutorial_Activities\ST2_Week_5_Tutorial_Activities> & C:\Users\LENOVO\AppData\Local\Python\pythoncore-3.14-64\python.exe c:/Users/LENOVO/Desktop/ST2_Week_5_Tutorial_Activities/towerofhanoi.py
Enter number of disks: 3
Move disk 1 from A to B
Move disk 2 from A to C
Move disk 1 from B to C
Move disk 3 from A to B
Move disk 1 from C to A
Move disk 2 from C to B
Move disk 1 from A to B
Total number of moves: 7
PS C:\Users\LENOVO\Desktop\ST2_Week_5_Tutorial_Activities\ST2_Week_5_Tutorial_Activities>

```

Task 4



Sierpinski Triangle



Enter an order:

5

Display Sierpinski Triangle

